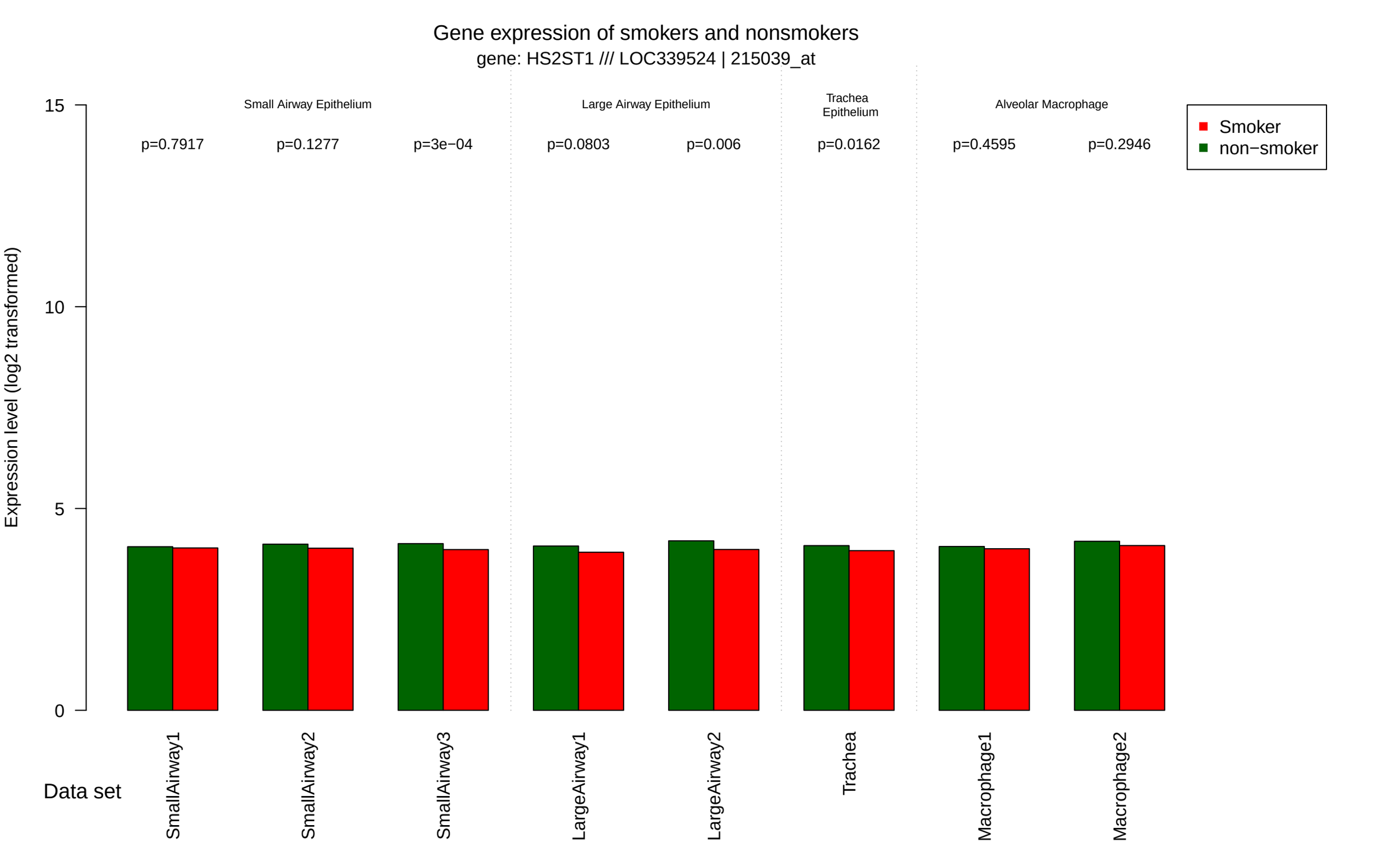
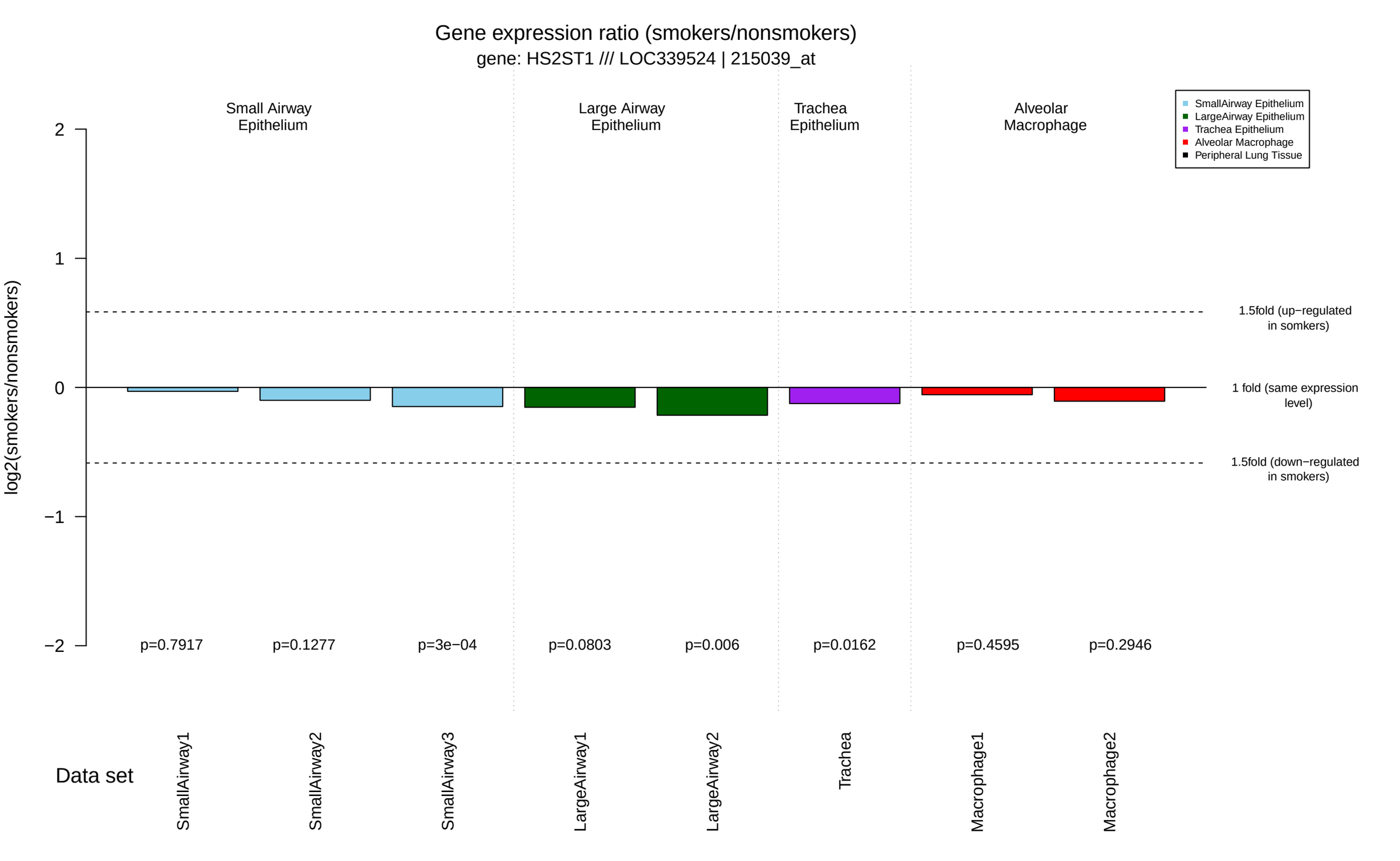
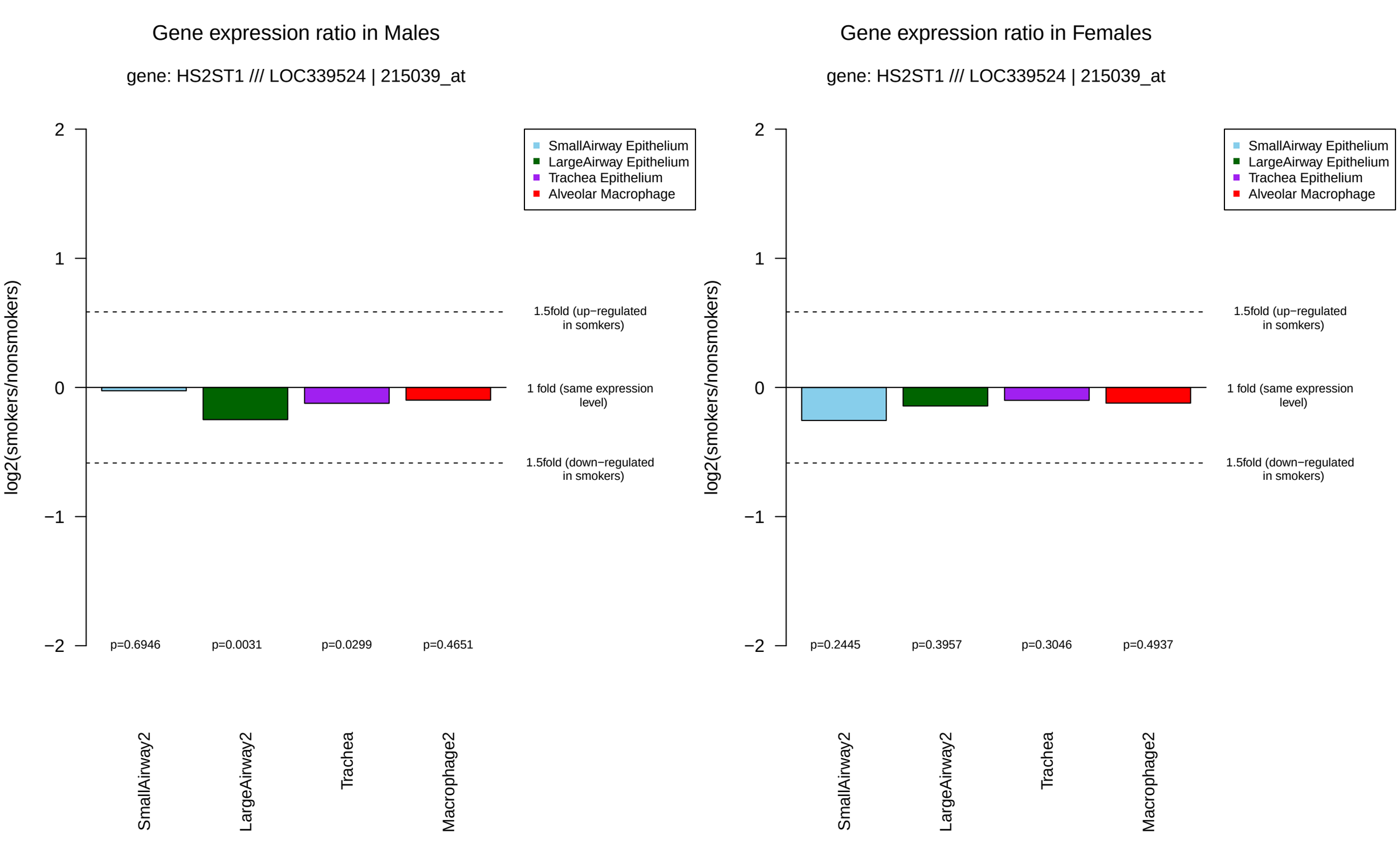
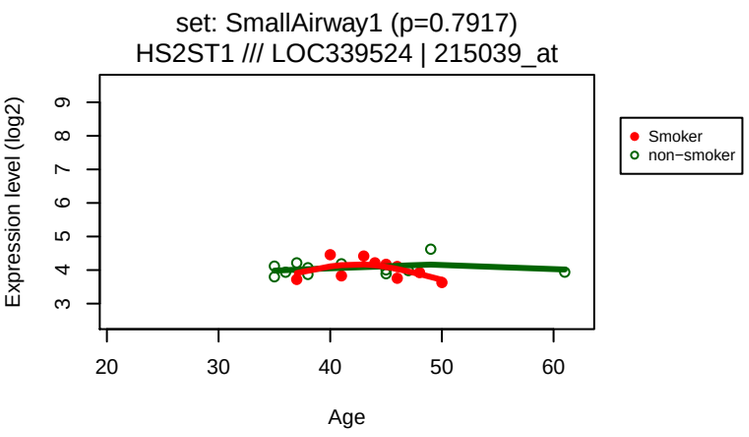
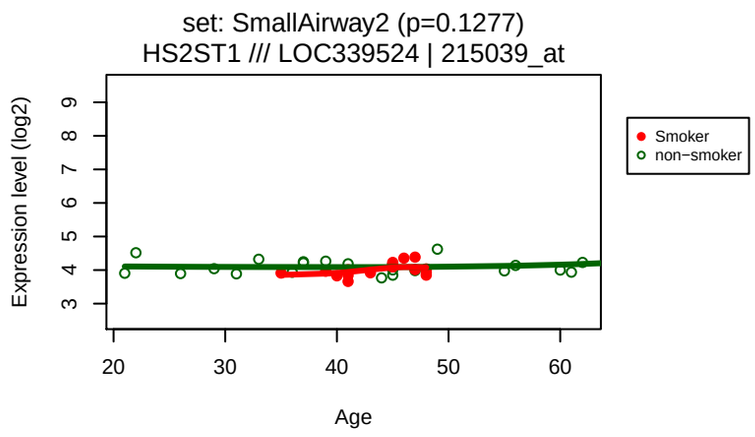
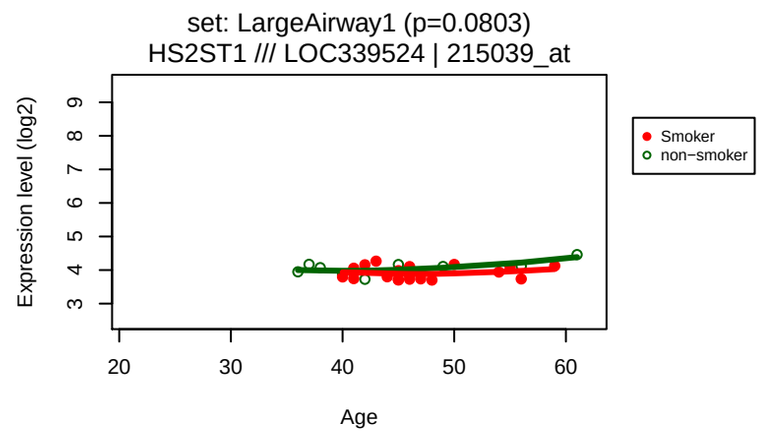

Gene expression of smokers and nonsmokers
gene: HS2ST1 /// LOC339524 | 215039_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7917
p=0.1277
p=3e−04
p=0.0803
p=0.006
p=0.0162
p=0.4595
p=0.2946
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

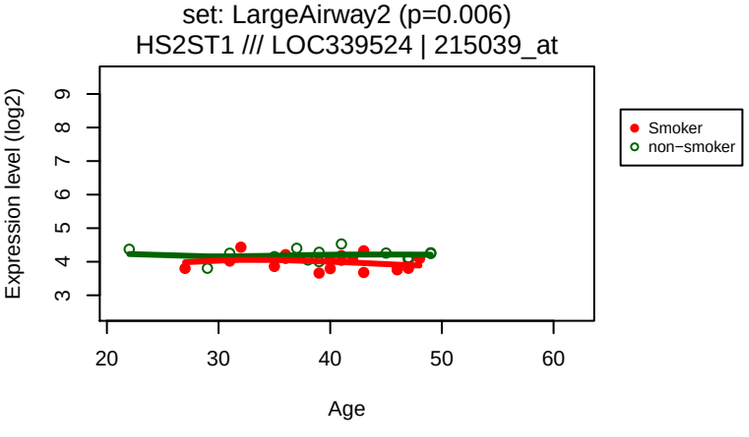

Gene expression ratio (smokers/nonsmokers)
gene: HS2ST1 /// LOC339524 | 215039_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7917
p=0.1277
p=3e−04
p=0.0803
p=0.006
p=0.0162
p=0.4595
p=0.2946
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

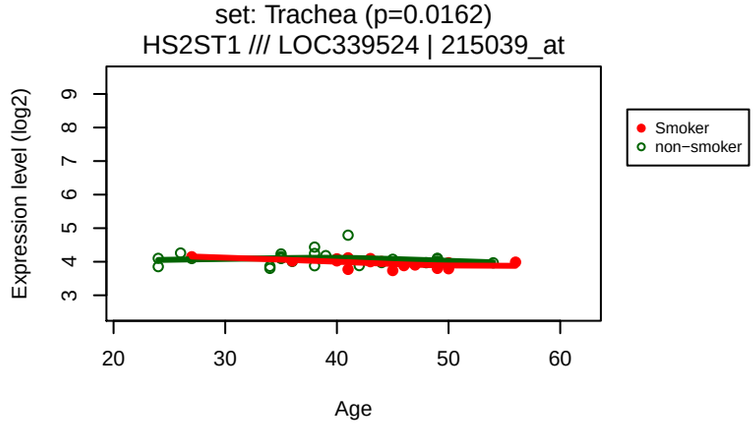

Gene expression ratio in Males
gene: HS2ST1 /// LOC339524 | 215039_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6946
p=0.0031
p=0.0299
p=0.4651
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HS2ST1 /// LOC339524 | 215039_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2445
p=0.3957
p=0.3046
p=0.4937
SmallAirway2
LargeAirway2
Trachea
Macrophage2

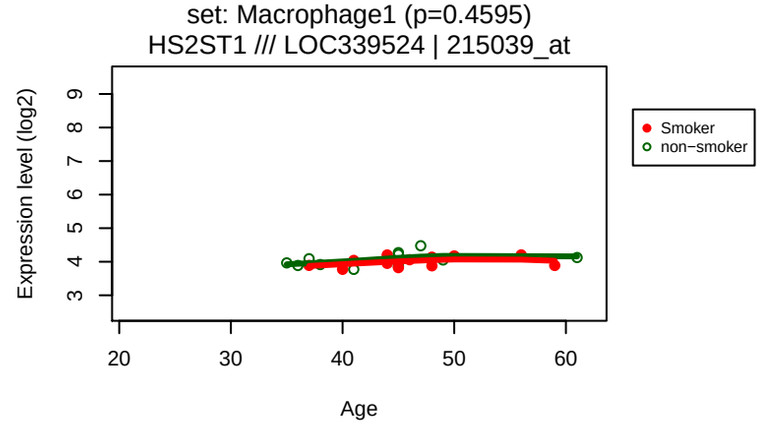

set: SmallAirway1 (p=0.7917)
HS2ST1 /// LOC339524 | 215039_at
Expression level (log2)
Age
Smoker
non−smoker

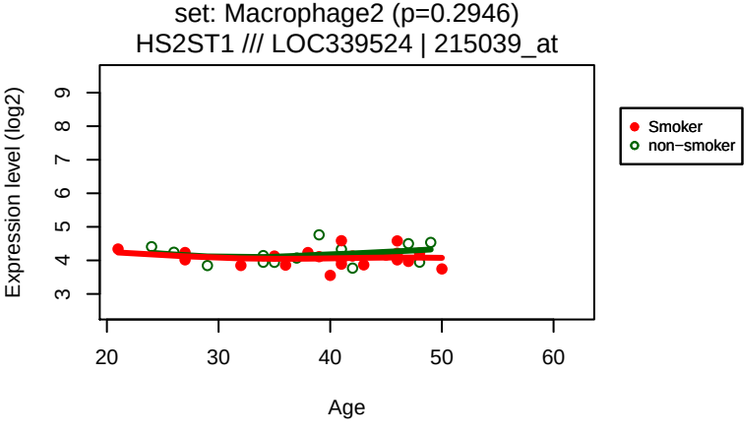

set: SmallAirway2 (p=0.1277)
HS2ST1 /// LOC339524 | 215039_at
Expression level (log2)
Age
Smoker
non−smoker

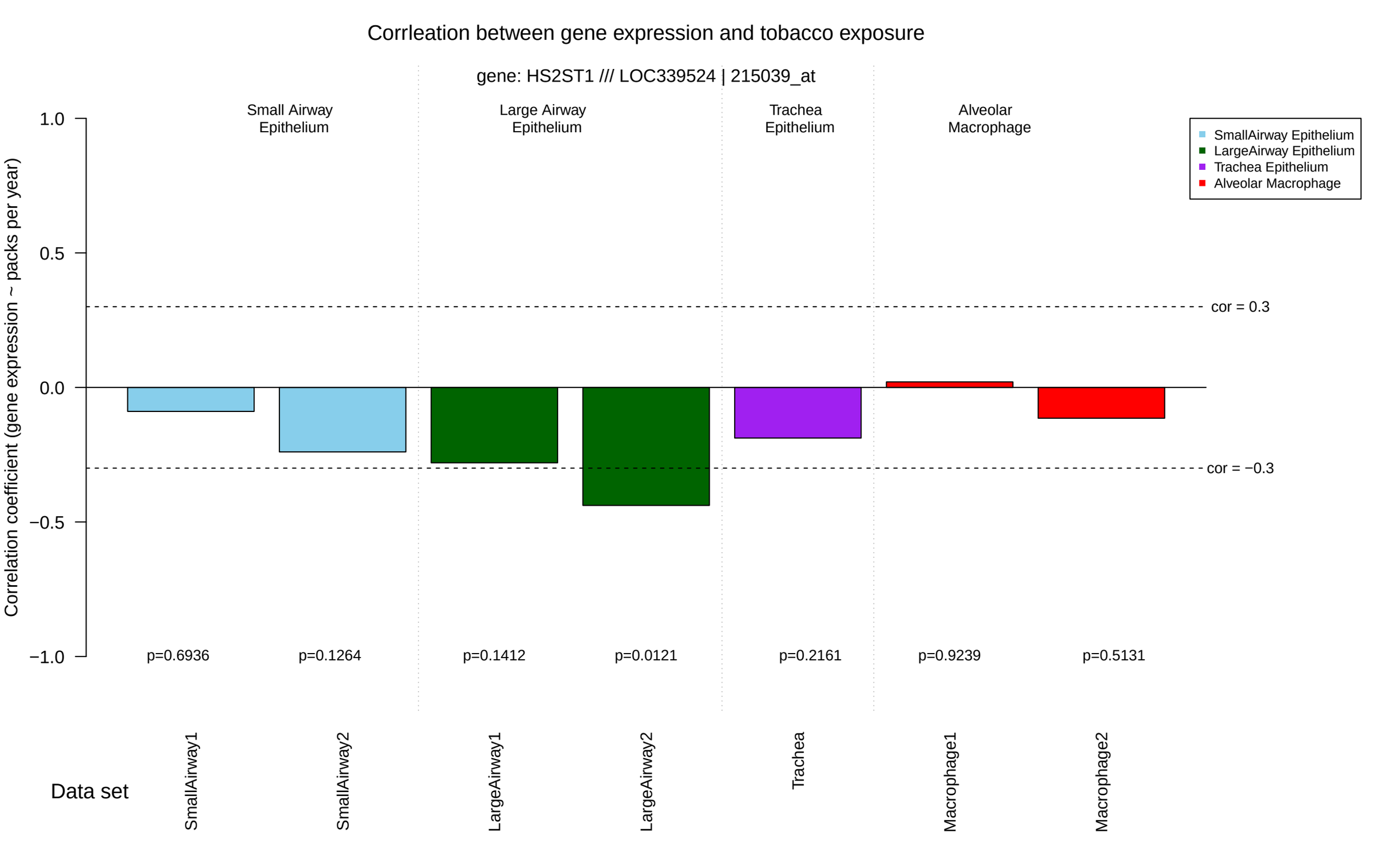

set: LargeAirway1 (p=0.0803)
HS2ST1 /// LOC339524 | 215039_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.006)
HS2ST1 /// LOC339524 | 215039_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0162)
HS2ST1 /// LOC339524 | 215039_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4595)
HS2ST1 /// LOC339524 | 215039_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2946)
HS2ST1 /// LOC339524 | 215039_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HS2ST1 /// LOC339524 | 215039_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6936
p=0.1264
p=0.1412
p=0.0121
p=0.2161
p=0.9239
p=0.5131
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2